

Wind power to methanol: Renewable methanol production using electricity, electrolysis of water and CO₂ air capture



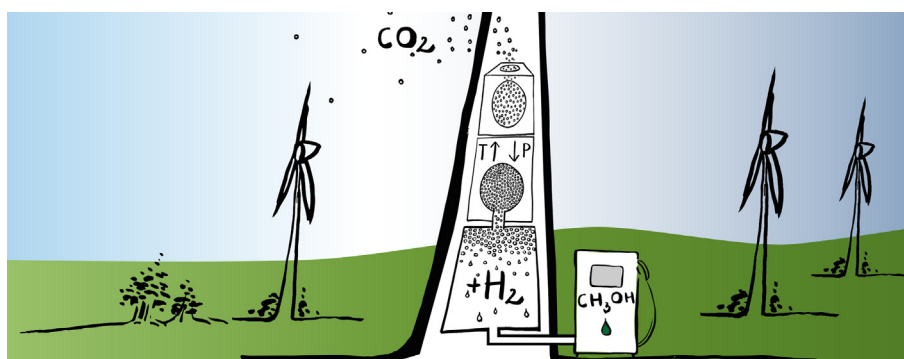
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HIGHLIGHTS

- Conversion of 100 MW wind power to methanol at 50% efficiency.
- Feedstock H₂O and CO₂ are captured from air.
- Design of air capture installation and methanol reactor based on experimental work.
- 10 MW wind turbines can accommodate the air capture installation in the base tower.
- Estimated methanol price: 800 euro per ton.

GRAPHICAL ABSTRACT



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ABSTRACT

A 100 MW stand-alone wind power to methanol process has been evaluated to determine the capital requirement and power to methanol efficiency. Power available for electrolysis determines the amount of hydrogen produced. The stoichiometric amount of CO₂ – required for the methanol synthesis – is produced using direct air capture. Integration of utilities for CO₂ air capture, hydrogen production from co-harvested water and methanol synthesis is incorporated and capital costs for all process steps are estimated.

Power to methanol efficiency is determined to be around 50%. The cost of methanol is around 300€ ton⁻¹ excluding and 800€ ton⁻¹ including wind turbine capital cost. Excluding 300 M€ investment cost for 100 MW of wind turbines, total plant capital cost is around 200 M€. About 45% of the capital cost is reserved for the electrolyzers, 50% for the CO₂ air capture installation, and 5% for the methanol synthesis system. The conceptual design and evaluation shows that renewable methanol produced from air captured CO₂, water and renewable electricity is becoming a realistic option at reasonable costs of 750–800 € ton⁻¹.

1. Introduction

The CO₂ concentration in the atmosphere increased from 250 ppm in the pre-industrial era to more than 410 ppm nowadays [1]. A large part of the increase can be contributed to use of fossil fuels. A decrease in CO₂ emission by reduction in fuel use is not expected. It is predicted

that the world energy use will only increase in the years to come [2]. Nowadays, it is widely accepted that increased CO₂ concentrations in the atmosphere lead to changes in the world's climate. Therefore, energy production with low CO₂ emissions is gaining interest. The installed power of renewable energy sources such as, wind power, solar PV and hydro-power are increasing yearly. A recent publication by

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Zappa et al. [3] showed that a 100% renewable power system is possible in Europe in 2050. However, implementation of large scale renewable energy requires storage methods for electricity [3–5].

For electricity storage over longer time periods – for example seasonal storage – conversion of electricity to chemicals is, next to hydro-power, an excellent method to store electricity [6,7]. One of the chemicals to store electricity is methanol [8]. The major advantage of methanol over hydrogen and methane is, that methanol is liquid at ambient conditions and thereby easy to store and transport. Furthermore, it is shown that methanol can be produced at higher overall efficiency compared to other chemicals [9–11]. Methanol is a very versatile product as it might be transformed back in electricity, used in gasoline engines or upgraded to diesel substituent dimethylether (DME). Moreover, methanol produced from captured CO₂ using renewable electricity is an excellent renewable carbon source for the chemical industry [9,12,13]. Using the methanol-to-olefins (MTO) process, olefins can be produced from methanol and by reforming to syn gas basically any chemical is within reach [14].

This current paper evaluates the standalone conversion of air and wind power to renewable methanol, using air capture to provide CO₂ and H₂O. A literature overview of issues regarding production of methanol from CO₂ and renewable energy is given by Gonzalez-Aparicio et al. [15]. Although the idea of methanol production from (wind) power and captured CO₂, including an economic evaluation thereof, has been presented in literature [9,16–22], none of these studies addresses integration options for utilities between electrolysis, CO₂ and H₂O capture and methanol synthesis. In this study options for heat integration between process steps are evaluated. This study is the first to combine the direct air capture of CO₂ and water with the production of methanol using renewable energy sources. The production of methanol from air and nuclear energy has been proposed by Steinberg and Dang [23] in 1977 already. Moreover, this study is one of the few [23,24] to include CO₂ capture in the process evaluation instead of assuming the availability of a pure CO₂ feed stream [21,16,22]. In these earlier studies, opportunities for process integration are disregarded. Besides integration of utilities, this is the first study to evaluate physical integration of process steps for standalone methanol production. The current studies assesses the complete integration of the process equipment in the base tower of the wind turbine.

1.1. Electrolysis

For electrolysis of water three main technologies are available today; alkaline electrolysis, proton exchange membrane (PEM) electrolysis and Solid Oxide electrolysis. An overview of the technologies is given in Table 1. In alkaline electrolysis the electrodes are immersed in a liquid electrolyte separated by a diaphragm. Normally, the electrolyte is a KOH-solution and OH[−] ions are passing through the diaphragm, releasing hydrogen at the cathode and oxygen and water at the anode. Alkaline electrolyzers are already commercially available at MW scale [25].

With proton exchange membrane electrolysis protons are passing through a membrane. Protons and oxygen are produced at the anode, the protons pass through the membrane and combine to hydrogen at

the cathode. Because of the corrosive acidic conditions noble metals are required for the electrodes, increasing the capital cost of the PEM electrolyser [25].

Solid oxide electrolyzers are operated at 700–900 °C, resulting in a lower energy requirement for hydrogen production, because of thermal activation. Therefore, with good heat integration, these electrolyzers are expected to be more efficient than alkaline and PEM electrolyzers. Currently, solid oxide electrolyzers are in the development phase and not commercially available yet [25].

1.2. Direct air capture

Recently, Sanz-Perez et al. [30] published a review comparing different methods for air capture of CO₂. Azarabadi and Lackner [31] gave an overview of different sorbents for CO₂ air capture. A short overview of available technology is given below. The group of Lackner at Arizona State University works on so-called moisture swing adsorption of CO₂ from air. The CO₂ is adsorbed in dry conditions and desorbed by increasing the humidity of the gas [32,33].

The Jones research group at Georgia Tech focuses on development of amine-based sorbents. The work includes amine impregnated [34,35] and grafted [36,37] on silica and metal organic frameworks [38]. The company Global Thermostat is affiliated with the group of Jones and uses to monolith bonded amines to perform direct air capture [39].

The group of Keith at the Harvard University develops an alkaline hydroxide process. CO₂ is captured by a reaction with liquid NaOH or KOH to form a carbonate solution [40]. The carbonate solution reacts with calcium hydroxide to form calcium carbonate (CaCO₃). At the same time the alkaline hydroxide is regenerated and can be reused in the absorber. The calcium carbonate is regenerated in a calcination loop at high temperature, a disadvantage of this process. Carbon Engineering develops the alkaline hydroxide/calcium calcination technology on commercial basis. A 1Mt-CO₂/year plant is designed in their latest paper [41]. The levelized cost per ton of CO₂ ranged from 94 to 232 dollar.

Steinfeld's group at ETH Zurich also investigated solid amine sorbents. The process is based on a temperature-vacuum swing adsorption to produce CO₂ from air [42]. The process is able to produce high purity CO₂ from air [43]. The company Climeworks is started by alumni of Steinfeld's group and is using granulated material with amines to capture CO₂ by a temperature vacuum swing adsorption. Earliest estimation of the cost of air capture are about 600 dollar per ton of CO₂ [44].

In this paper technology developed by the group of Brilman at the University of Twente is used. The commercial solid amine sorbent Lewatit VP OC 1065 is used to perform air capture of CO₂. This shows a unique feature of this work as the design is based on previous experimental work on adsorption [45] and desorption [46,47].

1.3. CO₂ to methanol conversion

Conventional methanol production technology is widely discussed in books by Olah et al.[48] and Bertau et al.[49] and therefore not further discussed here. A short overview of advanced methods for shifting the equilibrium of the methanol reaction – and thereby reducing the amount of unreacted material in the recycle – will be given. For example, the equilibrium can be shifted by the use of membranes as studied by van der Ham et al. [50], by Struis et al.[51] and by Gallucci et al. [52] showed increase in conversion and selectivity to methanol by the use of a zeolite membrane. Rahimpour et al. [53] showed 4.7% increase in the per pass methanol yield by the use of palladium-silver membrane tube walls.

Another solution to surpass the equilibrium conversion is to use a solvent to adsorb reaction products. Hagihara et al. [54] showed a methanol yield of 95% by the use of n-dodecane as extraction solvent. Westerterp et al. [55] showed almost full single pass conversion by

Table 1

Key performance indicators of electrolysis technologies [25–29].

	Alkaline	PEM	Solid Oxide	Unit
Temperature	60–90	50–80	600–1000	°C
Pressure	1.05–30	10–200	1–25	bar
Energy consumption	4.5–6.6	4.2–6.6	3.7–3.9	kWh m _{H₂} ³
Max. Capacity	5.3	1.1	n.a	MW
Capital cost	800–1500	900–2200	> 2000	€ kW ^{−1}
Technology readiness level (TRL)	9	5–7	3–5	–

absorption of products by TEGDME. Also Krishnan et al. [56] showed improved conversion up to 80% by the use of TEGDME as solvent although conversion rates were 2–3 times lower due to diffusion limitations. More recently, Xu et al. [57] showed improved conversion by the use of alcohol as a solvent. Adsorption of product vapors by silica-alumina powder with conversion up to 100% was shown by Westerterp et al. [58,59].

By operating at high pressure, the products are condensed and removed from the reaction phase, thereby shifting equilibrium. Condensation at high pressure was shown by van Bennekom et al. [60] at 200 bar and by Tidona et al. [61] at pressures up to 360 bar. High pressure operation was shown experimentally by Gaikwad et al. [62] and by Raymond et al. [63] at very high pressure (> 450 bar).

On the other hand, condensation can be initiated at lower pressures by a reduction in the (local) temperature. Haut et al. [64] showed increased conversion by a radial temperature gradient over the catalyst bed. Perko et al. [65] used a temperature gradient by a parallel hot and cold plate.

In the presented design the conversion of CO_2 to methanol will be performed in the Liquid-Out Gas-In Concept (LOGIC) reactor as introduced in previous work [66]. Using a temperature gradient to shift the reaction to the product side. Moreover, the outcomes of the modelling and experimental work on the LOGIC reactor as presented previously [67] are used.

1.4. Design basis

The Dutch government is planning several wind parks on the coast of the Netherlands with total installed power of around 10GW [68]. The (planned) locations of these wind parks have an installed power around 1.5GW per location. As design basis for this study it is assumed that 100 MW (6.7%) of this 1.5GW is available to produce methanol for 8000 h per year. Although the current design basis is related to Dutch situation, it should be realized that the technology selection and design presented in this work is independent of the location.

The aim of this paper is to give a first indication about the cost of standalone renewable methanol production and equipment sizes based on conceptual design of these experimentally proven technologies.

2. Blackbox evaluation

Prior to a more detailed process evaluation, a blackbox calculation is performed on feedstock prices for methanol production. Using the prices for CO_2 and H_2 the price of methanol can be calculated. Note that this first calculation only includes feedstock prices and includes no other OpEx nor CapEx costs. The results of the blackbox calculation for the methanol price, as a function of CO_2 and H_2 prices, is shown in Fig. 1. To show the economical operation window, the methanol sales price (375 € ton^{-1} , [69]) is shown by the dashed line. As renewable methanol might be sold by a premium price, the price of renewable methanol is assumed at 550 € ton^{-1} .

Prices for fossil hydrogen range from 1,500 € ton^{-1} for coal gasification and autothermal reforming of natural gas to 2,000 € ton^{-1} for steam methane reforming of natural gas [70]. However, to produce hydrogen in a renewable and anthropogenic carbon-free way, hydrogen should be produced by electrolysis of water or from biomass resources. According to literature [70], prices of hydrogen produced from biomass are in the same range as for fossil based resources. For example, hydrogen produced by biomass gasification (1,700–2,100 € ton^{-1}) [70], bio-photolysis by algae (2,100 € ton^{-1}) [70], hydrothermal gasification of wet biomass (3,400 € ton^{-1}) [71] or by fermentation (2,600 € ton^{-1}) [70]. Hydrogen produced by electrolysis is somewhat more expensive, as estimates range from 4,000 to 10,000 € ton^{-1} . [70].

Considering the objective of standalone methanol production using wind energy, biomass schemes are excluded and only electrolysis of water for hydrogen production is considered in this evaluation.

Consequently, the price of hydrogen is strongly correlated to the electricity price. Therefore, the effect of the electricity price is shown as the second x-axis in Fig. 1, assuming an electrolysis efficiency of 65%. Running a profitable operation of methanol production while using water electrolysis is, currently, very challenging. At current electricity prices of 0.07–0.10 € kWh^{-1} [72], the cost of produced hydrogen is too high to produce methanol at a competitive price compared to fossil fuel based methanol. At an electricity price of 0.03–0.05 € kWh^{-1} , the cost of CO_2 should be between 50–100 € ton^{-1} . In all cases, the capital and operational expenditures should be low to sell the methanol at a competitive price.

The cost of CO_2 produced by air capture, ranges from 100 to 1000 € ton^{-1} in literature [73,74]. Recently, a detailed paper of the system by Keith et al. [41] showed costs in the range of 93–232 $\text{\$ ton}_{\text{CO}_2}^{-1}$ already. A price of 200 $\text{€ ton}_{\text{CO}_2}^{-1}$ combined with (very) cheap hydrogen would potentially lead to a profitable production of methanol. Using CO_2 from a post-combustion capture would improve the range of profitable operation even more. The cost of CO_2 from a post-combustion capture is estimated in the range of 30–60 $\text{€ ton}_{\text{CO}_2}^{-1}$ [75,76]. In this study CO_2 is produced by air capture.

It can be concluded that, at this moment, the production of hydrogen by electrolysis is the main cost driver. Lower cost hydrogen, produced during periods of excess production of electricity, or available from biomass resources will improve the business case. With ongoing development on hydrogen production technologies, direct air capture and increasing availability of low-cost renewable electricity, the production of methanol from air, water and renewable electricity will become more and more competitive.

3. Process description

In the proposed process methanol is produced from CO_2 captured from air and H_2 produced by electrolysis of co-harvested water using renewable electricity. The design basis is 100 MW of electric input and 8000 operating hours per year. The schematic flow diagram of the process is shown in Fig. 2. First, it is determined how much hydrogen can be produced by the electricity provided. Next, the required stoichiometric amount of CO_2 for the methanol reaction is determined. The separations in the process are assumed to be 100% efficient. Co-adsorbed water during air capture can be used in the electrolyser and the co-produced water during methanol synthesis is recycled. The required amount of extra water – or excess of water – is shown by the magnitude of stream 13.

The main heat consuming steps in the process are the desorption of CO_2 (Q_{CO_2} in Fig. 2) and the reboiler of the distillation column (Q_{reb} in Fig. 2). When operating the electrolyser at elevated temperature, the heat produced during the production of hydrogen (Q_{H_2} in Fig. 2) can directly be integrated with the heat consuming steps in the process. If the heat demand exceeds heat production by electrolysis, further heat integration can be done by using the cooler and condenser duties. However, the temperature level of these heat duties is too low for direct heat integration. Therefore, the temperature level of these heat sources is – if required – upgraded by heat pumps, at the cost of spending electric energy and reduction in overall electric power to methanol efficiency.

The feed compressor of CO_2 and H_2 for the methanol reactor, the air fans and – if required – the heat pumps consume electric energy, at the expense of electricity available for hydrogen production. Therefore, the mass and energy balances are solved iteratively to make sure the total electric energy demand of the process sums up to 100 MW.

3.1. Electrolysis

Because of the significant lower capital cost and the mature state of alkaline electrolysis technology, alkaline electrolysers are used in this evaluation. The operating conditions for electrolysis, as presented in

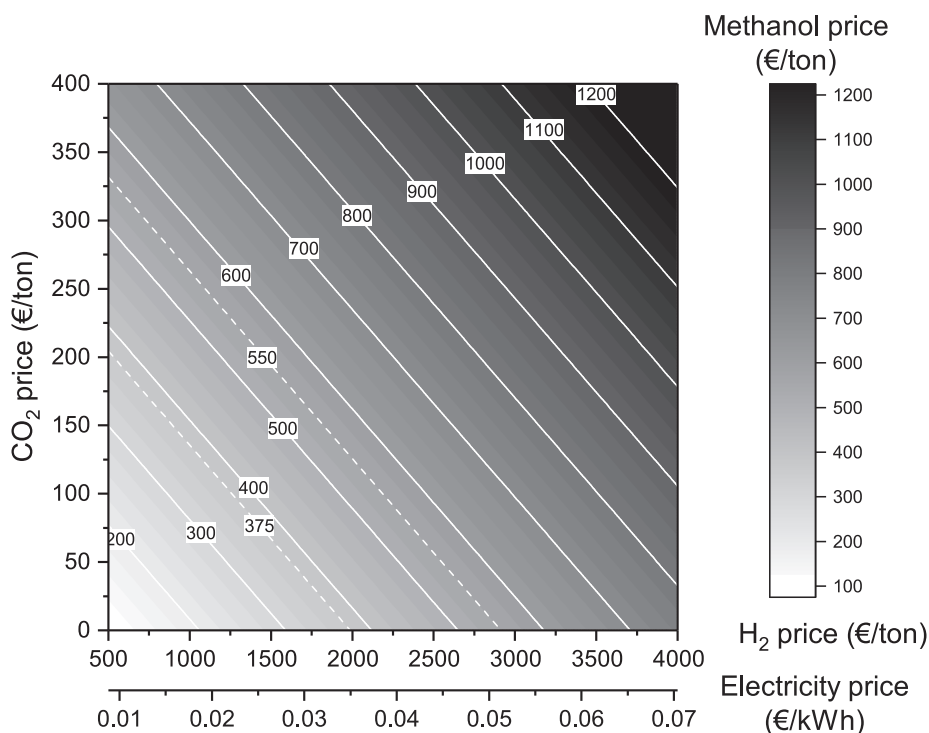


Fig. 1. Cost of methanol as a function of the feedstock cost. Apart from feedstock cost, no other OpEx or CapEx cost included. Efficiency of electrolysis is assumed 65% for electricity requirement. Dashed lines show methanol sales price (375 € ton⁻¹, [69]) and renewable methanol premium price (550 € ton⁻¹).

Table 2, are 120 °C and 30 bar. By operating the electrolysis at higher temperature opportunities for heat integration in the process are increased. Especially, heat integration for the energy required for desorption of CO₂ would reduce energy requirement of the overall process. Operating at 30 bar lowers the compression energy of hydrogen for the methanol synthesis as pressurizing liquid water requires less energy compared to gas compression.

3.2. CO₂ Air capture

For direct air capture, the solid amine technology developed at the

university of Twente is used. Fixed bed operation is used for the simplicity of operation as discussed in previous work [45]. Furthermore, because of the long contacting times of the sorbent with air to reach sufficient saturation, an system with moving sorbents is disadvantageous. The length of the bed is chosen such to minimize energy requirement for contacting, as discussed in previous work [45]. The air capture installation is sized on the stoichiometric amount of CO₂ required for methanol synthesis, based on the amount of hydrogen produced. The sorbent working capacities for CO₂ and water adsorption are determined from the isotherm values. For CO₂, the Toth isotherm using Bos parameters as presented in previous work [46] is used. For

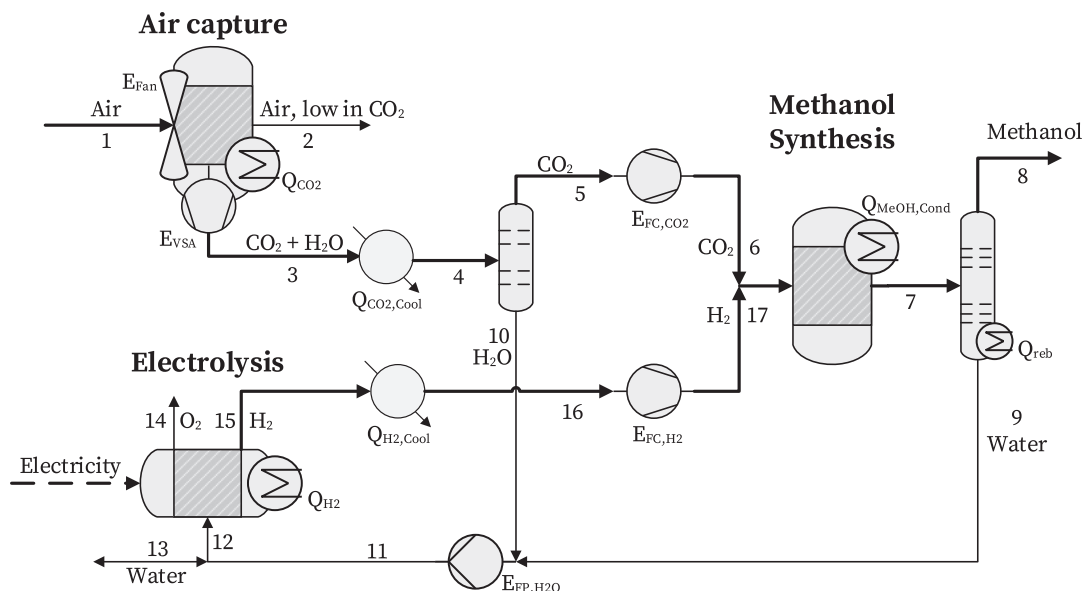


Fig. 2. Schematic process flow diagram, including electrolysis section, air capture section and methanol synthesis. Streams: (1) air inlet, (2) air outlet, (3) & (4) Capture CO₂ and H₂O, (5) & (6) CO₂, (7) Outlet methanol reactor, (8) Methanol product, (9–12) Water to electrolysis, (13) Purge water, (14) O₂, (15–17) H₂.

Table 2
Overview of assumptions for process evaluation.

Overall			Electrolysis				
E _{input}	100	MW	T _{elec.}	120	°C		
C _{wind turbines}	3000 [79]	€/kW	P _{elec.}	30	bar		
Methanol synthesis [67]			E _{req.}	5.0	kWh m ⁻³ _{H₂}		
			η _{elec.}	0.65	–		
P _{MeOH}	50	bar	C _{elec.}	1000 [25]	€/kW		
T _{cat,in}	240	°C	Distillation				
T _{cond,out}	60	°C					
F _{Prod}	40	mol/kg _{cat} /h	T _{feed}	80	°C		
F _{Q,cond}	5.5 × 10 ⁶	W/kg _{MeOH}	x _{distillate}	0.999			
F _{A,cond}	640	m ² /kg _{MeOH}	x _{bottom}	0.001			
F _{A,hex}	700	m ² /kg _{MEOH}	l _t	0.5	m		
C _{catalyst}	15	€/kg	d _{wt}	5	mm		
CO ₂ air capture	1: Base Case	2A: Low T _{ads}	2B: High T _{ads}	3A: Low P _{des}	3B: High P _{des}	4: Low oper. hours	Unit
Oper. hours	8000	8000	8000	8000	8000	4000	h/y
T _{ads}	15	5	25	15	15	15	°C
RH _{ads}	75	75	75	75	75	75	%
t _{ads}	2700	2700	2700	2700	2700	2700	s
v _{air}	0.3	0.3	0.3	0.3	0.3	0.3	m s ⁻¹
P _{des}	500	500	500	100	1000	500	mbar
T _{des}	115	115	115	115	115	115	°C
t _{des}	900	900	900	900	900	900	s
η _g & η _{s,CO₂}	0.55 [45]	0.55 [45]	0.55 [45]	0.55 [45]	0.55 [45]	0.55 [45]	0.55 [45]
η _{s,H₂O}	0.57	0.57	0.57	0.57	0.57	0.57	0.57
L _{bed}	0.03 [45]	0.03 [45]	0.03 [45]	0.03 [45]	0.03 [45]	0.03 [45]	0.03 [45]
D _{bed}	5	5	5	5	5	5	5
C _{sorbent}	30	30	30	30	30	30	€/kg
Sorbent stab.	2	2	2	2	2	2	years

the water capacity the Guggenheim, Anderson, de Boer (GAB) isotherm is fitted on experimental data by Veneman [77], see Fig. S1 in the Supporting Information.

The conditions for air capture are shown in Table 2. The average temperature in the Netherlands is 15 °C with a relative humidity (RH) of 75%. While the relative humidity is more or less constant around the year, the temperature has more variation. Therefore, cases at lower (5 °C) and higher (25 °C) temperature are defined. As presented in work by Bos et al. [47], to reduce energy requirement the desorption temperature should be higher than 100 °C. The pressure during desorption is 500 mbar for the base case. The effect of desorption pressure is investigated by calculating cases at 100 and 1000 mbar. The effect of desorption temperature on the working capacity is similar to the effect of desorption pressure and therefore not discussed separately. For more details on the effect of desorption temperature the reader is referred to previous work [78,47].

Using the stoichiometric time approach as presented by Yu et al. [45] the time for adsorption can be coupled to the gas and solid efficiency. The gas efficiency shows the amount of CO₂ removed from the air passing through the adsorbent bed. At 100% gas efficiency all the CO₂ in the air is removed. The solid efficiency shows the approach of the working capacity to the equilibrium working capacity. At 100% solid efficiency, both the adsorption and desorption equilibrium capacities are reached, resulting in a maximum working capacity.

The total sorbent mass is found by the required amount of CO₂ and the effective working capacity. The number of adsorbent beds is found by the total sorbent mass required and the mass per adsorbent bed. An adsorbent bed is assumed to have a 5 m diameter and 0.03 m length. The length of the sorbent bed is found by optimization of energy requirement using experimental values in the work by Yu et al. [45]. In the Supporting Information more detailed equations for the calculations performed are given.

3.3. Methanol section

The conversion of CO₂ to methanol will be performed in the Liquid-Out Gas-In Concept reactor as introduced in previous work [66]. The methanol reactor is scaled to the adiabatic reactor model as discussed in the concept optimization section of our latest work [67]. The conditions of 240 °C at the catalyst inlet, 60 °C at the condenser outlet and 50 bara are chosen, see Table 2. By normalizing the calculated heat flow and heat exchanger area to the mass of methanol produced, the numbers can be used to estimate sizes for an upscaled reactor.

The distillation column has been sized by the Fenske-Underwood method [80] to determine the minimum reflux and theoretical number of stages. The reflux ratio is assumed to be 1.2 times the minimum reflux ratio and the number of stages is determined by the graphical Erbar-Maddox approach. Column diameter is determined by the vapour flow and the column height is found by the number of stages and the height of one stage [81].

All the assumptions taken for the process evaluation are presented in Table 2. In the Supporting Information more details about the calculations performed and cost estimation of equipment are given.

4. Results and discussion

4.1. Base case

First, the detailed results of the base case are discussed, followed by a sensitivity analysis on the effect of the adsorption temperature, desorption pressure and operating hours. The base case will produce 65kton of methanol yearly, with 100 MW of electric energy input. This results in a power to methanol efficiency of 51%, as shown in Table 3.

In the base case, 87 MW is available for electrolysis resulting in 28 MW of heat production at an electrolysis efficiency of 65%. Since

Table 3

Main performance indicators for the five case defined in Table 2. A Lang factor of 5 is taken to calculate capital cost.

Parameter	1: Base Case	2A: Low T_{ads}	2B: High T_{ads}	3A: Low P_{des}	3B: high P_{des}	4: Low oper. hours	Unit
$E_{Electrolysis}$	87	90	82	82	86	87	MW
η_{P2M}^1	51	53	49	49	51	51	%
$\Phi_{m,MeOH}$	65	67	62	62	65	33	kton y^{-1}
Φ_{m,CO_2}	90	93	85	85	89	45	kton y^{-1}
T_{ads}	15	5	25	15	15	15	$^{\circ}C$
T_{des}	115	115	115	115	115	115	$^{\circ}C$
P_{des}	500	500	500	100	1000	500	mbar
$x_{CO_2,des}$	0.16	0.21	0.12	0.13	0.25	0.16	–
RH_{des}	25	23	26	5.2	45	25	%
Δq_{CO_2}	0.56	0.70	0.42	0.71	0.40	0.56	mol kg^{-1}
Δq_{H_2O}	2.8	2.6	3	4.9	1.2	2.8	mol kg^{-1}
$m_{sorbent}$	454	375	580	340	635	454	ton
N_{beds}	1428	1184	1824	1072	2000	1428	–
$\Phi_{v,air}$	6316	5222	8064	4727	8829	6316	$m^3 s^{-1}$
Q_{H_2}	28	29	27	27	28	28	MW
Q_{CO_2}	44	38	52	42	46	44	MW
E_{Fan}	3.7	3	4.7	2.8	5.1	3.7	MW
E_{VSA}	1.6	1.3	2.1	7.3	0.12	1.6	MW
$E_{Heat pump}$	6.3	4.2	9.6	6.4	7.2	6.3	MW
$C_{Electrolysis}$	87	90	82	82	86	87	M€
$C_{CO_2 capture}^2$	95	77	121	89	124	95	M€
$C_{Methanol}^3$	11	11	10	10	11	11	M€
C_{Total}^4	194	181	216	183	222	194	M€
$C_{CO_2 capture}^5$	308	254	403	360	353	414	€ $ton_{CO_2}^{-1}$
C_{Total}	298	268	350	298	344	596	€ ton_{MeOH}^{-1}
$C_{Total, incl. WT}^6$	758	713	836	785	809	1520	€ ton_{MeOH}^{-1}

¹ Power to methanol efficiency.

² Includes capital cost for sorbent, air fans, vacuum compressor and reactor steel.

³ Includes capital cost for feed compressors, methanol reactor and distillation.

⁴ Excludes 300 M€ capital cost for 100 MW of wind turbines.

⁵ Cost of CO_2 , including capital cost, cost for electric energy ($0.1€ kWh^{-1}$) and cost steam ($25€ ton^{-1}$).

⁶ Includes 300 M€ capital cost for 100 MW of wind turbines.

electrolysis is operated at elevated temperature, the heat produced can be integrated with the CO_2 capture system. However, 44 MW of heat is required for the desorption of CO_2 for the base case. Therefore, 16 MW of heat has to be integrated with lower temperature heat sources (e.g the cooler and condenser duties) using heat pump technology. Because heat pumps consume electricity, it is clear that by reducing the heat demand of the capture process the overall power to methanol efficiency can be increased.

The heat requirement for CO_2 desorption is shown in more detail in Fig. 3. The energy required for sensible heat of the sorbent takes about 45.6% of the total heat requirement. Recovery of sensible heat in fixed bed operation is difficult. At maximum, a cold bed finished adsorption and a hot bed finished desorption can be levelled in temperature, reducing the sensible heat by 50% at maximum heat exchange efficiency. This would reduce the heat pump duty by around 3.5 MW electric, boosting process efficiency. In literature [82] a heat integration concept between fixed beds using heat pump technology was shown to reduce energy requirement. Further recovery of the sensible heat of the sorbent, could be performed by circulating the sorbent between separate adsorption and desorption reactors. This would enable counter-current heat exchange between sorbent leaving and entering the desorption reactor, increasing the maximum amount of heat exchanged. Moreover, this would save the sensible heat of the reactor, reducing energy requirement by 2.5 MW (5.7%). Application of full heat exchange

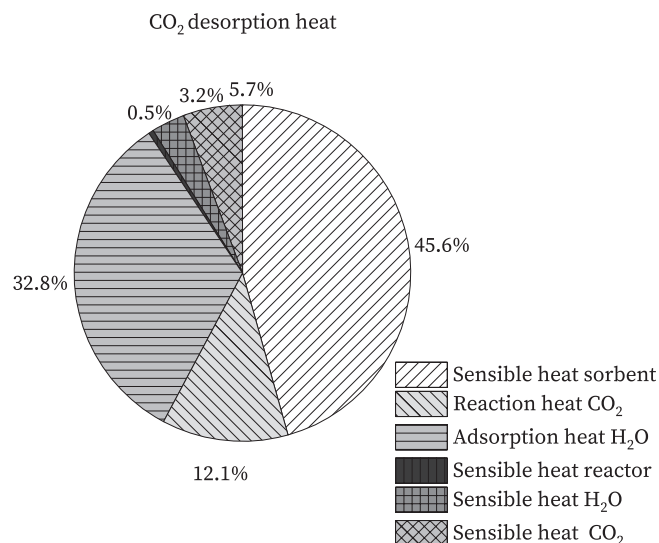


Fig. 3. Heat consumption for CO_2 desorption. Total heat requirement 44 MW or 14 GJ/ton.

combined with a circulating system would take away the need of a heat pump. However, systems with sorbent circulation add complexity to operation of the system.

Another method of reducing sorbent sensible heat requirement is increasing the solid efficiency. Currently, only 55% percent of the equilibrium working capacity is used. Increasing solid efficiency by increasing the adsorption time, would reduce gas efficiency due to reduced uptake rates when the sorbent bed approaches equilibrium. A method to further increase solid efficiency while maintaining gas efficiency could be reduction of the air velocity with increasing bed loading.

Further reduction of heat required for desorption can be done by reducing the amount of water co-adsorbed. Weather conditions can change and pre-conditioning of air is too costly, as a few thousand cubic meter of air are processed per second (see Table 3). To reduce the water working capacity, the amount of water desorbed could be reduced by increasing the relative humidity during desorption. Using steam for desorption would increase relative humidity, however as shown in previous work [47] this does not result in a nett reduction of energy requirement.

Furthermore, by changing the absolute pressure during desorption the maximum relative humidity in the reactor is influenced. As relative humidity is the ratio between the partial pressure of water and the saturation pressure, reducing absolute pressure will reduce the maximum relative humidity in the reactor. Therefore, based on water working capacity, the desorption pressure should be increased. However, for CO_2 working capacity, the pressure should be reduced. Therefore, the desorption pressure should be optimized to the ratio of CO_2 and water desorbed. The optimal ratio in CO_2 and water working capacity would be around two, as nett two hydrogen molecules are required for one CO_2 molecule to produce methanol.

The total electricity requirement of the process is shown in Fig. 4. As shown, currently the heat pump is accountable for about 50% of the electricity requirement. Reduction of heat pump duty is already discussed above. The electricity requirement of the air fans can be reduced by either reducing the pressure drop over the adsorbent bed or by increasing the gas efficiency. Because already a very short bed is used, further reduction of pressure drop is unlikely. Changing a bed of sorbent particles for a monolith type of adsorbent material might open opportunities for further pressure drop reduction. As already pointed out above, increasing gas efficiency might be done by dynamic operation of the air fans. By optimizing the air velocity as a function of the bed loading an increased gas efficiency and optimized pressure drop

Process electricity consumption

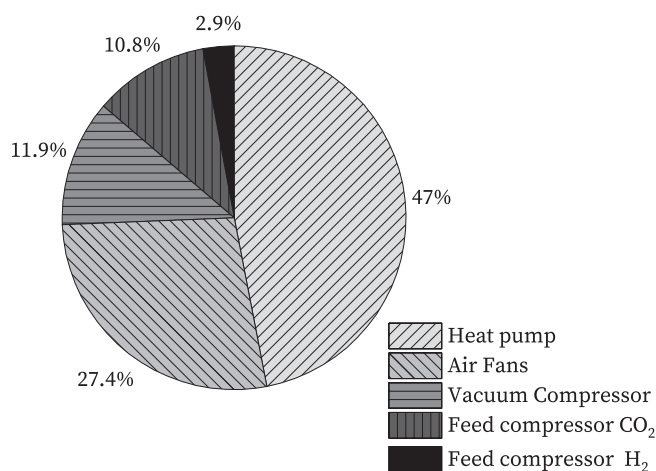


Fig. 4. Electricity consumption in the process apart from the electrolysis. Total electricity consumption is 13 MW.

might be achieved.

The total capital cost for the power to methanol plant is estimated around 200 M€, this excludes a investment of 300 M€ for 100 MW of wind turbines. The capital cost split-up – as shown in Table 3 – of the plant is roughly 45% for the electrolyzers, 50% for the CO₂ capture installation and 5% for the methanol section including purification.

The capital cost of the CO₂ capture installation is dominated by the sorbent cost, as shown in Fig. 5. The cost for the sorbent is assumed 30 € kg⁻¹, which is 50% of the current – lab scale – price. Further reduction of price can be expected as soon as CO₂ capture is applied at large scale. The minimum price is expected to be price of polystyrene (around 5 € kg⁻¹). The sorbent lifetime is estimated to be two years and thus approximately 16,000 cycles. An increase in sorbent lifetime would significant reduce the capital cost. Therefore, it is important to further investigate sorbent lifetime and stability. Increasing the working capacity of the sorbent would reduce the required sorbent mass and reduce capital cost. At the same time, sensible heat requirement will be reduced. Therefore, it is important to increase sorbent efficiency and working capacity. By reducing the amount of sorbent mass required, the number of adsorbent beds is reduced too. This will

Capital cost air capture

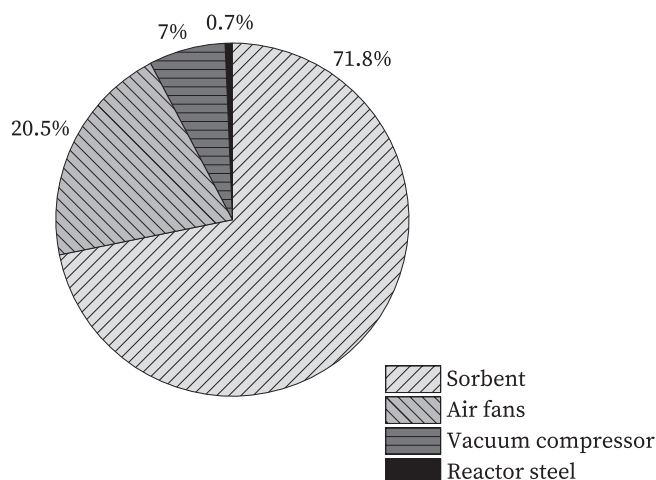


Fig. 5. Capital cost for the air capture installation, total capital cost for air capture installation is 95 million euro.

Capital cost methanol

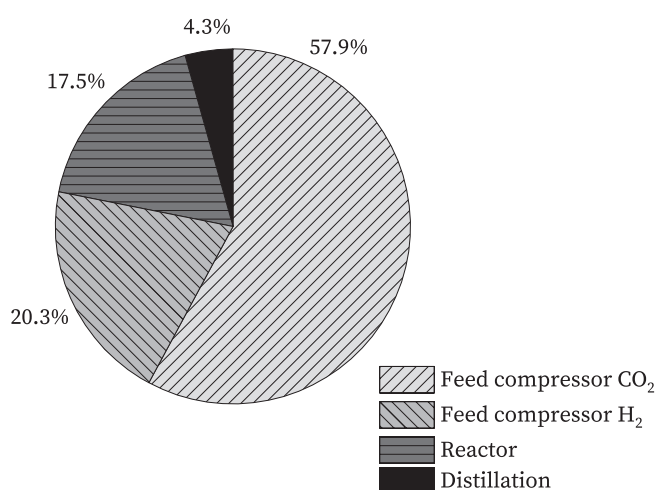


Fig. 6. Capital cost for methanol synthesis, including compression and distillation. Total investment cost for the methanol section is 11 million euro.

further reduce the capital costs further, as seen by the capital cost of the fans in Fig. 5. Per bed one fan is required, consequently reducing the number of beds will reduce cost. It can be concluded that the capital cost of the CO₂ capture installation is mainly a function of the sorbent working capacity and sorbent lifetime.

The cost of the methanol reactor system including purification is given in Fig. 6. The cost of the system is dominated by the cost of the feed compressors for the methanol reactor. The capital cost of the feed compressors is a function of the required power. Reducing the operating pressure of the methanol reactor will decrease investment cost. However, this will reduce reactor productivity and therefore increase capital cost of the reactor. The cost of the methanol reactor system can therefore be optimized by the ratio in capital cost for the feed compressors and the reactor.

4.2. Adsorption temperature

Comparing the case 1 and 2 in Table 3 it is seen that reducing the adsorption temperature increases working capacity. In the discussion above, it was hypothesised that increasing working capacity would increase power to methanol efficiency. In Table 3 it is confirmed that the 5 °C has the highest efficiency. By increasing the working capacity the desorption heat requirement is reduced, increasing overall efficiency.

As shown in Table 3, the number of beds is reduced, due to the changed working capacity. In reality however, the productivity of the system will be changed as the number of beds is fixed by design. On the other hand, it can be decided to put part of the capture system in standby in the case of increased productivity. In this way, system energy efficiency will be increased although capital cost will suffer. The number of adsorbent beds range from 1,000 to 2,000. At first sight, this seems to be an enormous number of beds. By assuming a total bed height of 0.5 m including overhead space, the total height of the CO₂ capture system is 1000 m. However, in order to produce 100 MW of electric wind power, at least 10 wind turbines are required with a height of more than 150 m each. Therefore, easily 200 beds can be incorporated per tower and the complete CO₂ capture system could be fitted in the base towers of the ten wind turbines required. In other words, each wind turbine can accommodate the required air capture installation for methanol production, in line with its own power generation capacity.

It is seen that the productivity is increased at lower adsorption

Table 4
Production cost for fossil, biomass and electrolyse based methanol.

		Methanol technology	Carbon source	Capacity (kton/y ⁻¹)	Cost (€ ton ⁻¹)
Fossil	Blug et al. [83]	Syn gas	Natural gas reforming	1,800	200
	Blug et al. [83]	Syn gas	Coal gasification	1,800	250
Biomass	Carvalho [84]	Syn gas	Forest residu gasification		600
	Balegedde Ramachandra [85]	Syn gas	Reforming of glycerol		500
Electrolysis	Hank et al. [21]	Direct CO ₂	Captured CO ₂ (outside scope)	4–10	610–1450
	Perez-Forbes et al. [16]	Direct CO ₂	Captured CO ₂ (outside scope)	800	720
	Mccord et al. [24]	Direct CO ₂	CO ₂ flue gas capture by calcium carbonate	365	1,400
	Rivera-Tinoco [22]	Direct CO ₂	CO ₂ (input)	16	900
	This work	Direct CO ₂	CO ₂ air capture by solid amines	65	800

temperatures, although in previous work [46] it is shown that the adsorption kinetics are significantly reduced at low temperature. However, the effect of kinetics on solid and gas efficiency is not included in the current evaluation. As both solid and gas efficiency are negatively influenced by reduced kinetics, the advantage of operating at low adsorption temperature might be decreased when kinetics are included in the evaluation. Therefore, more detailed modelling of the adsorption bed and process is required to fully optimize the operating conditions.

The total fixed capital investment is strongly correlated with the working capacity of the sorbent, as shown in Table 3. This is a result of the increase in mass of sorbent required with lower working capacity. Moreover, because the number of beds is higher with a low working capacity the capital cost of the air fans is increased. Therefore, capital investment can be optimized by increasing the CO₂ working capacity for the sorbent.

4.3. Desorption pressure

With decreasing desorption pressure, the working capacity of CO₂ is increased as shown in Table 3. However, at the same time the water working capacity is significantly increased because of the reduced relative humidity at low absolute pressure. Therefore, the effect on power to methanol efficiency and cost of methanol production are not the same compared to changes in lower adsorption temperature.

Due to increased water working capacity and lower absolute pressure the duty of the vacuum compressor is increased, see Table 3. This is a consequence of increased flow and pressure difference over the compressor. Accordingly, the power available for electrolysis is reduced resulting in a lower methanol production. The duty of vacuum compressor might be reduced by (partial) condensation of the water upstream of the vacuum compressor reducing the flow through the compressor. Due to the increase in CO₂ working capacity, the total capital cost of the system is reduced because of the significant reduction in sorbent requirement.

On the other hand, operation of the desorption at atmospheric pressure reduces CO₂ working capacity and increases capital cost. Because of increased sorbent inventory, the energy requirement of desorption is increased increasing the cost per ton of methanol. Again, this shows the importance of optimizing working capacity for both CO₂ and water.

4.4. Operating hours

For the cases discussed, 8000 h of operation is assumed. Since the wind velocity and thus power output is part of varying weather conditions this might be optimistic. In this study we assume that only 6.7% of the maximum power output of the wind park is used. Therefore, even at low wind velocity the energy output will be high enough to produce methanol. Nevertheless, a fourth case is defined with reduced operation hours, see Table 2. As seen in Table 3, 50% reduction in operating hours will double the price. The plant is still designed for 100 MW of

electricity and therefore capital cost is equal to 8000 h of operation. However, the methanol output is halved due to reduced operating hours. Thereby a significant increase in price is seen. This shows the importance of including site-specific partial load and operating hours in future more detailed techno-economic evaluations.

4.5. Methanol production

The total cost per ton of methanol of around 800 € ton⁻¹ (see Table 3) is about 1.5 times the estimated premium price of renewable methanol of 550 € ton⁻¹. This opens opportunities for renewable methanol production, especially with decreasing prices for wind energy and electrolysis [86,28]. It should be noted that the numbers presented are first estimate capital costs and do not include operational cost such as cost for employees and maintenance. Estimated values for future cost are a reduction of 30% in capital cost for wind turbines [86] in 2030, 500 €kW⁻¹ for electrolysis [28] in 2030 and the cost of air captured CO₂ is expected to go down to 100 € ton⁻¹CO₂ [74]. With these numbers, methanol could be produced at cost around the estimated premium price for renewable methanol.

The production cost for methanol production based on fossil fuels, biomass based and direct conversion of CO₂ by electrolysis based hydrogen are shown in Table 4. The production cost of fossil based methanol is in the range of 200–250 € ton⁻¹, depending on the feedstock. Biomass or electrolysis based methanol is not able to compete with fossil methanol purely on production cost. Despite cost reduction for renewable technologies, as discussed above, competing with fossil based fuels is extremely difficult. When using fossil fuels, both the energy and carbon source, are recovered from the earth at low cost. Switching to electrolysis and CO₂, serious investment have to be made to harvest energy (e.g. wind turbines or solar PV) and recovery of the carbon source by CO₂ capture. With the growth, harvesting and transport of biomass also significant cost are involved, raising the production price of methanol. Moreover, fossil fuels can benefit from the economy of scale, as the plants for fossil fuel based production are significantly larger compared to renewable based plants in Table 4. For carbon neutral methanol production the various options should not only be compared on an economic basis, but also using Life Cycle Analysis. Such comparison of alternatives is however beyond the scope of this work.

4.6. Outlook

Governmental legislation on emissions and ambitions for a climate-neutral future do open up opportunities for renewable methanol based processes. The new Renewable Energy Directive II (RED II) of the European Union requires 14% of renewable energy, including methanol, to be used in the transport sector by 2030. In total 66 countries have put targets and mandates for the implementation of biofuels, some as high as 27% [87]. Moreover, general climate targets of the European Commission foresee a climate-neutral European Union in 2050 [88].

However, the chemical industry still requires a carbon source for the production of chemicals such as plastics. Next to biomass, renewable methanol – or indirectly CO₂ – is an excellent feedstock for the chemical industry. Processes for the production of olefins from methanol are already commercially available and by conversion of methanol to syngas, practically any chemical is within reach.

5. Conclusion

A 100 MW wind power to methanol plant has been evaluated based on energy requirement and capital cost. A power to methanol efficiency of around 50% has been found. The cost of methanol is 300€ ton⁻¹ excluding and 800€ ton⁻¹ including wind turbine capital cost. Excluding 300 M€ investment cost for 100 MW of wind turbines, total plant capital cost is around 200 M€. About 45% of the capital cost is reserved for electrolysis, 50% for the CO₂ air capture installation, and 5% for the methanol synthesis system. It is shown that heat produced by electrolysis can be fully integrated with the heat required for desorption of CO₂. Additionally, it is proven that the air capture installation, electrolysis equipment and methanol reactor can be installed in the base tower of the wind turbine, in line with its own power generation.

Apart from the efficiency of electrolysis, the main parameter to be optimized is the sorbent working capacity for the CO₂ air capture system, which is also weather and thereby location dependent. Increasing working capacity would reduce both, the heat demand and the capital cost. Both, heat demand for CO₂ desorption and capital cost for CO₂ capture are dominated by the mass of sorbent required in the process. Reducing heat demand for CO₂ would increase process efficiency and reduction in sorbent mass reduces cost per ton of methanol. Moreover, optimizing the co-adsorption of water to the requirement for methanol synthesis can further optimize system efficiency and cost.

Keeping in the mind the extra process steps required to produce renewable methanol, a first estimate of 800 € ton⁻¹ is a good starting point. With expected cost reduction for renewable electricity, electrolysis and direct air capture the production of renewable methanol is promising as electricity storage medium and feedstock for the chemical industry.

CRediT authorship contribution statement

M.J. Bos: Writing - original draft, Visualization, Conceptualization, Formal analysis. **S.R.A. Kersten:** Supervision, Conceptualization. **D.W.F. Brilman:** Writing - review & editing, Supervision, Conceptualization, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.apenergy.2020.114672>.

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